

# QIANG GUO

PhD Applicant | AI for Finance, Operations Research, Statistics & Machine Learning

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## EDUCATION

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| <b>The Hong Kong University of Science and Technology (HKUST)</b> , Hong Kong<br><i>MSc in Electronic Engineering</i>          | <b>Sep 2018 – Nov 2019</b> |
| <b>Nanjing University of Aeronautics and Astronautics (NUAA)</b> , Nanjing, China<br><i>BEng in Aircraft Power Engineering</i> | <b>Sep 2014 – Jun 2018</b> |
| <b>National Tsing Hua University (NTHU)</b> , Hsinchu, Taiwan<br><i>Exchange Student, Mechanical Engineering (Fall 2016)</i>   | <b>Sep 2016 – Jan 2017</b> |

## RESEARCH INTERESTS

AI for credit risk and financial decision-making; causal inference and experimentation; machine learning under distribution shift and cold start.

## PROFILE

~7 years of applied modeling and decision-making work in consumer and SME credit at Ant Group, ByteDance and Du Xiaoman. Designs large-scale pricing experiments, builds and refits ML models for newly launched segments, works on network-based transaction modeling, and develops fraud detection models, all deployed in production credit systems. Seeks doctoral training to put this work on rigorous methodological foundations.

## RESEARCH-RELEVANT EXPERIENCE

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| <b>Ant Group</b> – Ant Consumer Finance (Huabei Installments), Beijing<br><i>Senior Expert, Risk Strategy; owns credit line and pricing strategy for a consumer installment portfolio</i> | <b>Dec 2025 – Present</b> |
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- **Experimental design & causal inference:** Designed an unbiased, statistically powered, platform-wide price-discount experiment to estimate segment-level discount ROI (lifecycle  $\times$  risk tier  $\times$  creditworthiness tier) and locate its inflection point. Used the results to build a risk-based price-increase policy for price-insensitive users that offsets margin lost to discounts. Defined requirements for price-response models.
- **Cold start for a new segment: data enrichment & operator identification:** Built the credit strategy for a new lending vertical (lithium-battery purchase financing for heavy-truck drivers), a relatively small segment the general product had not managed at a granular level. The segment lacked business-operation signals (identity, business cash flow, operating capability). Enriched the data by combining manufacturer-provided user data with internal risk data, identified self-employed/small-business operators, and mined features and refit models for the new population. The strategy raised approval rate and full-line coverage while holding risk stable.
- **Network-based modeling:** Modeled supply-demand networks to assess whether transaction needs are plausible in higher-risk offline merchant settings, and used this to raise full-line coverage only for selected transactions. Used AI to identify, generate and match network structures, run optimal model refits, and condense the results into deployable decision rules, which were then put into production.
- **AI-assisted research workflow:** Applied AI to feature mining for vertical segments and automated experiment data collection and monitoring.

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| <b>ByteDance</b> , Beijing<br><i>SME Credit Risk Strategy Expert (led teams of 2)</i> | <b>Dec 2022 – Aug 2025</b> |
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- **Decision modeling under risk constraints:** Designed a two-dimensional risk  $\times$  creditworthiness framework for credit-line decisions in SME lending, with channel-specific eligibility boundaries and incremental offer optimization. With risk constraints held, approval rate rose 50% (relative), credit lines doubled and risk fell 20%.
- In merchant lending, used risk tiering to allocate credit exposure by cycle, competitiveness and customer response. Segment risk fell 30% and outstanding balance doubled in one year.
- **Alternative data evaluation:** Ran ROI evaluations on 8 categories of external data (business registration, annual reports, merchant acquiring, e-commerce, ad spend, tax invoices, customs, owner-enterprise linkage) and onboarded them for SME profiling, creditworthiness and GMV estimation. Designed the test plans and wrote the evaluation reports.

Du Xiaoman Financial, Beijing

Oct 2021 – Dec 2022

*Risk Strategy Expert*

- **Response & sensitivity modeling:** Used high-value, response and price-sensitivity models to reallocate line and price offers to approved-but-undrawn customers, lifting utilization by 5%. Built an offline low-risk recall model for re-approving expired customers.
- **Field experiment:** Designed a credit-validity-extension experiment that moved risk screening to the transaction stage to address data staleness and adverse selection in recall. Risk fell 5% and utilization rose 10%.
- **Feature construction:** Built small-business signals from business registration and offline activity data and validated them against customer profiles and macro data. Initial credit lines rose 10% while relative risk fell 30%.

Du Xiaoman Financial, Beijing

Jun 2019 – Oct 2021

*Senior Algorithm Engineer*

- **Fraud detection models:** Developed 3 generations of telecom-fraud detection models from scratch, covering basic scam patterns, large-loan differentiation and real-time features. Reached 90%+ fraud recall and cut fraud incidence to 0.01%. The project won the company Innovation Award two quarters in a row.
- **Feature mining & imbalanced classification:** Mined 20+ highly predictive fraud features from device and app-list data, lifting online recall by 10%. Led a liveness-detection program that included fraud-risk features from liveness and OCR images; it doubled impostor-detection precision and cut manual review volume by 85% at 99% recall.
- **Data infrastructure:** Built a T+1 case-monitoring platform that stopped 2 large organized fraud rings, plus real-time feature services and data pipelines.

## SKILLS & CERTIFICATIONS

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**Programming & Data:** SQL, Python, Shell, Java, Hadoop, Spark (proficient: SQL, Python, Shell, Hadoop; familiar: Java, Spark)

**Methods:** Experiment design and A/B testing, ML modeling (feature engineering, IV analysis, model refit), risk modeling, network-based feature construction

**Certifications:** PMP

**Languages:** English (professional), Mandarin (native), Spanish (A1)